EU INNOVATION HUB

EU Innovation Hub for Internal Security

Annual Event Report 2024



EU INNOVATION HUB FOR INTERNAL SECURITY ANNUAL EVENT

in cooperation with the CERIS community

Brussels, 28 November 2024

EU INNOVATION HUB ANNUAL EVENT REPORT 2024

PDF | ISBN 978-92-9414-001-2 | DOI: 10.2813/0138653 | ISSN 2812-3077 QL-01-25-006-EN-N

Neither the European Union Agency for Law Enforcement Cooperation nor any person acting on behalf of the agency is responsible for the use that might be made of the following information.

Luxembourg: Publications Office of the European Union, 2025

© European Union Agency for Law Enforcement Cooperation, 2025

Reproduction is authorised provided the source is acknowledged.

For any use or reproduction of elements that is not under the copyright of the European Union Agency for Law Enforcement Cooperation, permission must be sought directly from the copyright holders.

Cite this publication: Europol (2025), EU Innovation Hub Annual Event report 2024, Publications Office of the European Union, Luxembourg.

This publication and more information on Europol are available on the Internet.

www.europol.europa.eu

Introduction

On 28 November 2024, the EU Innovation Hub for Internal Security (the 'Hub') held its Annual Event in Brussels.

The event gathered internal security practitioners from EU Member States, representatives from EU institutions and agencies, research and technology organizations, academic experts in the field of security, industry representatives and NGOs.

The Hub annual event was organised in cooperation with the Commission's Community of European Research and Innovation for Security (CERIS).

The central theme of the event, "Shaping responsible solutions for internal security", emphasized the need to develop effective innovative security solutions fostering internal security that at the same time align with citizens' expectations for privacy, transparency, and accountability.

The EU Innovation Hub for Internal Security is a cross-sectoral EU platform which aims to ensure coordination and collaboration between innovation actors in the wider field of internal security.

The Hub, composed of various EU Justice and Home Affairs agencies, the European Public Prosecutor's Office, the European Commission (DG HOME and DG JRC), and the General Secretariat of the Council including the EU Counter Terrorism Coordinator's Office, works to provide the latest innovation updates and effective solutions to support the efforts of internal security actors in the EU and its Member States, including justice, border security, immigration, asylum and law enforcement practitioners.

*For more information on the Hub, please **click here**.*

High-Level Opening



The co-chairs of the EU Innovation Hub Steering Group, Zsolt Szolnoki, Senior High Councillor, Ministry of Interior, Hungary, and Nicolas Bessot, Head of the Innovation and Security Research Unit, Directorate-General for Migration and Home Affairs, European Commission, opened the high-level session by underlining the strategic importance of the Hub and its relevance for end-users in the EU Member States. Mr Szolnoki emphasized the essential involvement of practitioners and industry collaboration in driving innovation activities, noting that while responses to a survey of Member States' research and innovation capabilities were mostly positive, some challenges and needs for enhancement remain. Mr Bessot highlighted the importance of internal security for competitiveness, acknowledging potential challenges while emphasizing shared objectives among stakeholders and praising the Hub's maturity and its ability to tackle priorities via its new cluster approach.

Opening keynote speeches were delivered by Bartjan Wegter, EU Counter-terrorism Coordinator, and Olivier Onidi, Deputy Director General, DG Migration and Home Affairs, European Commission.

Mr Wegter discussed the accelerating pace of technological change and the increasing complexity of security challenges, noting the blurred lines between traditional terrorist threats and other activities, such as state actors using terrorist tactics. Mr Wegter emphasized the critical role of innovation in addressing these challenges and highlighted the EU's longstanding investment in internal security innovation. He outlined four key focus areas for 2025: foresight and key enabling technologies, biometric recognition, AI for enhancing data analytics and predictive law enforcement, and ensuring LEAs have access to information exchanged by illicit actors. He emphasised the importance of EU independence and self-reliance in innovation, with a focus on both technology and people. The Draghi and Niinistö reports were referenced, calling for the Hub to be opened up to Member States for information exchange and continued engagement with industry. Finally, Mr Wegter underlined the necessity to accelerate the Hub's work, with a proposal to organise an internal security-focussed high level strategic workshop, aimed at strategic reflection on new technological solutions.

Mr Onidi addressed growing concerns about advanced technologies falling into the hands of criminals, over the past decade. He highlighted the importance of Europol's transformation into a fully operational law enforcement agency and the role of its Innovation Lab in enhancing operational capabilities. The Hub's potential in advancing legislation was noted, with its contribution to the EU AI Act's assessment. Mr Onidi emphasised the role of the Hub in facilitating AI technology uptake and the priority focus on encryption for the next two years, as well as the need to establish a dedicated group of experts on encryption to devise key technical requirements. Mr Onidi confirmed the Commission's support for a high-level event for decision-makers in EU internal security, as suggested by Mr Wegter.

Panel #1: Foresight as a Means to better prepare for Emerging Security Challenges

Introduction: Marta Cygan, Director, DG Migration and Home Affairs, European Commission
Moderator: Hans Martin Pastuszka, CERIS expert, Fraunhofer Institute
Panellists:

- Alexandra de Maleville, JRC, European Commission
- Thierry Hartmann, DCIS, French Ministry of Interior, AHEAD project coordinator
- Ricardo Neisse, Senior Research Officer, Frontex
- Mark Wittfoth, Head of Observatory, Europol Innovation Lab



The aim of this panel discussion was to analyse the key components that should be addressed for a more tailored foresight approach to civil security and to make the necessary links with Key Enabling Technologies (KETs). In addition, the discussion focused on ways to strengthen end-user's foresight capacities and better preparedness towards emerging security challenges in the EU stemming from KETs.

In her introduction, Marta Cygan emphasized the importance of foresight in anticipating evolving security challenges. She highlighted that a forward-thinking approach is crucial for building resilience against future crises. Understanding emerging uncertainties, opportunities, and megatrends is essential, especially in the context of increasing unpredictability. She underscored the commitment to embedding foresight and preparedness in EU policy, as emphasized by President Von der Leyen, to enhance the EU's resilience and autonomy. Ms Cygan also reflected on the lessons from the COVID-19 pandemic, where the failure to foresee supply chain vulnerabilities underscored the necessity of forward-looking policy-making.

The panel discussed the importance of foresight in law enforcement, with insights from a JRC studies that demonstrated how strategic foresight can be effectively conducted through literature reviews, horizon scanning, and Delphi surveys. Foresight helps identify new technological developments, understand their implications for threats and capabilities, and build scenarios that produce actionable roadmaps for decision-making. Examples included foresight on biometric technology and quantum technologies, highlighting the need for proactive planning to address future challenges. The ability to anticipate multiple future scenarios and develop roadmaps to achieve desirable outcomes was emphasized as essential for keeping citizens safe in a rapidly changing environment.

The discussion on key components highlighted the importance of having a broad view of technology and involving experts to map and analyse technological developments. Developing scenarios requires input from both technology and operational experts to ensure practical application. Understanding an organization's current position in technology development is crucial. The panel stressed the need for an interdisciplinary approach, involving a wide spectrum of stakeholders, and the importance of management buy-in for successful implementation. Training on foresight methodologies and the development of dedicated approaches were noted as vital steps to translate foresight into practical capability roadmaps. The ultimate goal is to bridge the gap between innovative technologies and their real-life applications in addressing security challenges.

Panel #2: Advancing Biometric Technology and Identity Management in the EU: Exploring the Need for Testbeds and Sandboxes

Introduction: Philippe Harant, Head of Strategy, Capabilities and Coordination Unit, eu-LISA
Moderator: Javier Galbally, Senior Capability Building Officer, Research and Development, eu-LISA

Panellists:

- Christoph Busch, Professor at the Norwegian University of Science and Technology (NTNU)
- Frøy Løvåsdal, Senior Advisor, National Police Directorate, Norway
- Sébastien Marcel, Senior Researcher at Idiap and Professor at the Université de Lausanne, Switzerland



Mr Harant opened the panel by highlighting the omnipresence of biometrics in our daily lives, from unlocking mobile phones to crossing borders and opening bank accounts. He emphasized the continuous growth of the global biometric technology market, which reached EUR 65 billion in 2023. He also pointed out the crucial role of biometric technologies in enhancing border security and process efficiency, but noted that the EU AI Act classifies most biometric applications as high-risk. He suggested that the EU should develop independent evaluation capabilities to maintain technological sovereignty, as alternative to reliance on the US NIST[1].

The panel discussion highlighted the critical role of biometric systems in the EU's internal security landscape, particularly for law enforcement and border management. Prof. Busch drew comparisons with vehicle testing, stressing the necessity of testing biometric recognition systems in real-life conditions, rather than relying on synthetic data. The major challenge identified was data availability for testing large-scale systems, with the BEAT[2] project's experience showing the need to move testing sites to where the data resides. The discussion underscored the absence of a legal framework allowing the use of real data for system testing, which is essential to ensure systems do not discriminate. While ISO has made strides in standardization, testing frameworks still need to be developed further to support comprehensive evaluation of biometric systems.

As to standardization and independence in testing, the discussion also addressed the paradox of AI systems, including biometric recognition, where the capacity to test for bias is limited despite widespread understanding of potential biases. The feasibility of using synthetic data for testing was debated, highlighting that while synthetic data can help assess certain system characteristics, it cannot replace real data entirely due to the diversity of real datasets that synthetic data cannot mimic. The panel emphasized the need for independent testing capabilities within the EU to ensure representative and unbiased evaluation of biometric technologies. Additionally, comments from the audience called for a reassessment of GDPR in the context of the EU AI Act, suggesting a separate legal framework for training and operational data processing.

[1] National Institute of Standards and Technology

[2] https://cordis.europa.eu/project/id/284989/reporting

Panel #3: The Benefits of Explainable AI in Internal Security

Introduction: Andrei Linţă, Deputy Executive Director Capabilities and Chief Information Officer, Europol
Moderator: Maximilian Zocholl, Specialist Technology & Innovative Solutions, Europol Innovation Lab

Panellists:

- Babak Akhgar, Director of CENTRIC, Sheffield Hallam University
- Irina Orssich, DG for Communications Networks, Content and Technology, European Commission
- David Reichel, Project Manager Justice, Digital and Migration Unit, EU Agency for Fundamental Rights
- Kelly Vink, Legal advisor AI & data science, National Police of The Netherlands



Mr Linţă's introductory remarks highlighted the significance of innovation at Europol, particularly in developing AI tools. He noted that AI systems can only benefit the internal security landscape if users can assess and explain the results produced by these systems. As AI systems become more complex, explaining their performance becomes more challenging. To ensure trust in these outcomes, it is crucial to audit the performance of AI systems, a necessity driven not only by technical requirements but also by ethical imperatives.

The panel discussed the paramount importance of compliance with fundamental rights when using AI systems, as a vital duty and not an optional consideration. The ability to interpret AI-supported decisions and ensure accountability was emphasized, with the burden of proof falling on those using AI. There was a focus on understanding the transparency required for different systems and recognizing that human investigators can also be biased. Balancing the need for AI explainability without compromising performance was debated, noting that the application of agreed principles through standardised procedures can contribute greatly to accountability. Additionally, the panel addressed the risk that AI compliance requirements could restrict the ability to combat crime effectively.

The panel further explored whether AI performance must be sacrificed for explainability, emphasizing that this depends on individual cases and specific conditions. Comparing machine bias with human bias, the discussion revealed that AI systems could perform tasks more accurately and with less bias than humans. The importance of testing AI systems in realistic scenarios was highlighted, as well as the necessity of strategic vision to determine which AI systems should be used. The need for AI literacy and a multidisciplinary approach to explainable AI was stressed, ensuring that users understand how AI systems operate. Lastly, it was noted that while those subject to AI systems do not need to understand entirely how they work, oversight bodies must possess the necessary knowledge to ensure proper usage and accountability.

Panel #4: Encryption & Access to Data

Introduction: Jorge Espina, Deputy National Member for Spain at Eurojust, and chairperson of the Eurojust Cybercrime Working Group
Moderator: Gwendal Le Grand, Deputy Head of the European Data Protection Board (EDPB) Secretariat

Panellists:

- Martijn Egberts, National Public Prosecutor for Digital Investigations, The Netherlands
- Anders Fällgren, Director of Division for Police Issues, Ministry of Justice, Sweden
- Tiphaine Romand-Latapie, Team Leader, Synacktiv Reverse Engineering



In his introduction, Mr Espina emphasised the impact of innovation, particularly encryption, on investigations at Eurojust. He highlighted that the judiciary often operates reactively, which poses challenges in the face of rapid technological advancements. Despite these challenges, progress must continue, and innovation in legislation, such as the mutual recognition of judicial decisions across the EU and efforts in digitalization and e-evidence, is crucial. Judicial authorities require tools to manage encryption under controlled conditions to access and read data during investigations, with EU Agencies providing necessary support that may be beyond the capabilities of individual Member States. The need for a balance between encryption for security and tools to ensure security despite encryption was underscored.

The panel discussed the importance of communication between lawyers and engineers, highlighting that while lawyers safeguard principles, engineers focus on practical solutions. Understanding what law enforcement authorities can do without judicial decisions is crucial for accessing essential data in investigations. The conversation often lies in bridging the gap between legal frameworks and technological capabilities. The lack of data access hinders evidence collection, which led to the establishment of the High-Level Group on access to data for effective law enforcement. The successful use of data from cases like EncroChat, in Sweden and several other Member States, was cited as an example of how data access can lead to significant convictions and violence prevention. While citizens value cybersecurity and privacy, there is also a need for lawful access to data to combat crime.

Referring to challenges and solutions in encryption, the discussion also covered the specific challenges posed by encryption in the EU, such as the complexity of mobile phone encryption and the difficulties in obtaining devices for research. Upskilling in this domain is challenging due to the high demand and competition for talent. The EU faces market price disparities compared to the US, affecting the competitiveness of Research & Development services. To remain competitive, it is crucial to hire, train, and retain talent within the EU. Additionally, the reliance on non-EU countries for services poses risks, as the work funded by the EU could potentially be used against it. The panel concluded that addressing these challenges requires a strategic approach to ensure technological sovereignty and effective law enforcement capabilities.

Conclusion



The Hub Steering Group co-chairs Zsolt Szolnoki and Nicolas Bessot concluded with a unified commitment to leverage innovation and collaboration to address future security challenges and ensure a secure environment for all EU citizens.

Key takeaways included the importance of developing innovative security solutions that align with citizens' expectations for privacy, transparency, and accountability. As the keynote speakers and panellists highlighted the necessity of a collective effort to tackle evolving security threats, there is an undeniable need for continuous collaboration, foresight, and technological advancement. The co-chairs called upon all actors to play their role in making Europe fit for the digital age, reinforcing the event's central theme of "Shaping responsible solutions for internal security."